

KRATI JOSHI

Backend Engineer | Node.js · TypeScript · PostgreSQL · Kafka

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PROFESSIONAL SUMMARY

Backend Engineer with 2+ years building scalable, production-grade systems for enterprise clients. At SAI Computers Limited, shipped 120+ REST APIs powering SMRITI 3.0 — an AI-analytics utility platform deployed across 4 government electricity DISCOMs. Deep expertise in Node.js, TypeScript, PostgreSQL, and Kafka. Proven in RBAC design, multi-tenant architecture, microservices, and AI-integrated backends.

EXPERIENCE

Backend Developer • SAI Computers Limited

Feb 2024 – Present

- 120+ production REST APIs:** Designed and maintained 120+ production REST APIs using Node.js, TypeScript & Express.js serving 4 government DISCOMs.
- RBAC & Auth:** Built JWT authentication + dynamic RBAC framework covering 8+ roles with granular feature-level permissions, achieving zero unauthorised-access incidents.
- Performance optimisation:** Cut average API response time by ~40% via in-memory caching strategies, PostgreSQL index tuning, and query rewrites on high-volume reporting endpoints.
- AI Analytics integration:** Integrated SaiGPT (NL-to-SQL) enabling non-technical business users to query operational data in natural language, reducing analyst turnaround from days to minutes.
- Event-driven pipelines:** Architected Kafka producer-consumer pipeline for asynchronous OCR image processing, handling 10,000+ field images/day with dead-letter queue failure handling.
- Multi-tenant backend:** Engineered a shared-schema multi-tenant system with middleware-enforced org-scoping, ensuring zero cross-tenant data leakage across concurrent enterprise users.

KEY PROJECTS

SMRITI 3.0 — Enterprise Utility Management & AI Analytics Platform

2024 – Present

Tech: Node.js · TypeScript · Express.js · PostgreSQL · SaiGPT · AWS S3

- Designed multi-tenant org management system adopted by PVVNL, DVVNL, Tripura & Meghalaya DISCOMs, supporting 1,000+ concurrent users across shared infrastructure.
- Built configurable Widget Framework: analysts configure custom SQL-backed KPI widgets without any developer intervention.
- Achieved ~35% faster dashboard load times via a cache-aside pattern with intelligent key-level invalidation on data mutations.
- Designed RBAC with feature-level granular permissions; zero cross-tenant data leakage verified by automated integration tests.

OCR & Image Processing Microservices Platform

2024

Tech: Node.js · TypeScript · Apache Kafka · PostgreSQL · AWS S3

- Built fully async OCR microservices pipeline processing 10,000+ meter & pole images/day from field operations.
- Implemented Kafka producer-consumer flow with retry logic (3x exponential backoff) + dead-letter topics; near-zero message loss in production.
- Reduced manual field-data entry by ~60% by automating OCR extraction and structured persistence, replacing a previously human-operated workflow.

TECHNICAL SKILLS

Languages & Runtime: Node.js · TypeScript · JavaScript · SQL

Frameworks & ORM: Express.js · Prisma · Joi · Tailwind CSS

Databases: PostgreSQL · MySQL · MongoDB

Architecture: REST APIs · Microservices · RBAC · JWT Auth · Multi-Tenant Systems · Kafka · API Security · System Design

EDUCATION

B.Tech — Computer Science (Artificial Intelligence)	2020 – 2024
AKTU (Dr. A.P.J. Abdul Kalam Technical University)	GPA: 8.01 / 10 (80.12%)
12th (Senior Secondary)	2020
UP Board	77%
10th (Secondary)	2018
UP Board	87.5%